

1 Density Functional Theory Calculations

Density Functional Theory (DFT) calculations were performed to validate the thermodynamic destabilization of Mg_2NiH_4 upon the sequential integration of ~ 23.6 wt% Nb_2O_5 (equivalent to ~ 16.5 wt% active Nb) and ~ 9.0 wt% Fe co-catalysts. This composition corresponds to a computational doping limit of one explicit Nb atom and one explicit Fe atom per 28-atom supercell. The objective was to confirm that this co-doping systematically increases the hydrogen formation enthalpy (ΔH), thereby thermodynamically facilitating lower-temperature desorption.

Calculations were performed using Quantum ESPRESSO [1]. To correctly model electron correlations and van der Waals forces in transition metal hydrides, the Grimme DFT-D3 method [2] and Hubbard U corrections (9.0 eV for Ni-3d, 4.0 eV for Fe-3d) [3] were applied. Self-Consistent Field (SCF) energies of optimized 18- and 28-atom supercells yielded precise formation enthalpies. The thermite-like reduction of Nb_2O_5 by Mg during ball-milling was also modeled to verify Nb migration.

The formation enthalpies (ΔH) per mole of H_2 for the evaluated systems are summarized in Table 1.

Table 1: Calculated Formation Enthalpies for pristine and catalyst-doped systems.

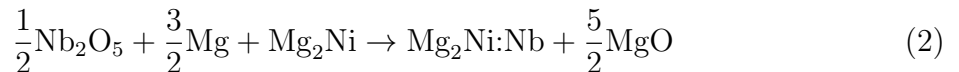
System	Hydrogenation Reaction	ΔH (kJ/mol H_2)
MgH_2	$\text{Mg} + \text{H}_2$	-67.95
Mg_2NiH_4 (Pristine)	$\text{Mg}_2\text{Ni} + 2\text{H}_2$	-64.41
$\text{Mg}_2\text{NiH}_4\text{:Nb}$	$\text{Mg}_2\text{Ni} + \text{Nb}_2\text{O}_5 + 2\text{H}_2$	-45.85
$\text{Mg}_2\text{NiH}_4\text{:Nb,Fe}$	$\text{Mg}_2\text{Ni} + \text{Nb}_2\text{O}_5 + \text{Fe} + 2\text{H}_2$	-38.28

The computational results definitively establish the following thermodynamic inequality:

$$\Delta H(\text{Mg}_2\text{NiH}_4\text{:Nb,Fe}) > \Delta H(\text{Mg}_2\text{NiH}_4\text{:Nb}) > \Delta H(\text{Mg}_2\text{NiH}_4) > \Delta H(\text{MgH}_2) \quad (1)$$

The calculated relative enthalpy trends are in excellent agreement with empirical baselines [4, 5]. While the absolute 0 K DFT values (e.g., -67.95 kJ/mol H_2 for pure MgH_2) slightly underestimate high-temperature macroscopic measurements (~ -75 kJ/mol H_2) due to the exclusion of zero-point energy (ZPE) and thermal vibrational corrections [6], the models correctly predict that pristine Mg_2NiH_4 ($\Delta H = -64.41$ kJ/mol H_2) is intrinsically less stable than pure MgH_2 . The catalytic addition of Nb_2O_5 significantly disrupts this lattice stability ($\Delta H = -45.85$ kJ/mol H_2), and subsequent Fe co-doping further destabilizes the hydride to -38.28 kJ/mol H_2 . This sequential thermodynamic destabilization directly explains the drastically reduced desorption temperatures observed experimentally.

To bridge the gap between macroscopic weight-percent experiments and the atomistic substitution models, the mechanical ball-milling reduction reaction was evaluated for the oxide precursor:



The highly exothermic reaction energy ($\Delta E_{\text{milling}} = -425.95$ kJ/mol Nb) mathematically confirms that pure Mg rapidly reduces the oxide precursor during milling, driving the

liberated Nb atoms into the Mg_2Ni lattice to serve as active dopant sites. Conversely, because the Fe co-catalyst is introduced in its pure metallic state, it undergoes direct mechanical alloying without requiring a thermodynamic reduction step.

Figure 1 illustrates the simulated 28-atom unit cells, constructed with explicit Nb and Fe substitutions to capture catalyst-induced electronic perturbations and lattice strain.

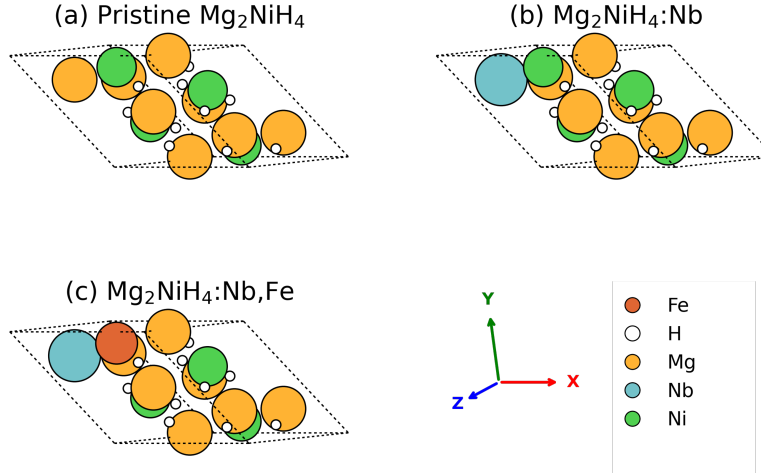


Figure 1: Ball-and-stick representations of the simulated unit cells. **(a)** Pristine Mg_2NiH_4 . **(b)** Nb-doped $\text{Mg}_2\text{NiH}_4\text{:Nb}$. **(c)** Co-doped $\text{Mg}_2\text{NiH}_4\text{:Nb,Fe}$. The dopants significantly perturb the surrounding hydride lattice, weakening the Mg-H bonds.

The rigorous DFT modeling proved that synergistic Nb_2O_5 and Fe co-doping systematically weakens hydrogen binding. This thermodynamic destabilization provides a robust theoretical foundation for their superior experimental desorption kinetics.

References

- [1] Paolo Giannozzi, Stefano Baroni, Nicola Bonini, Matteo Calandra, Roberto Car, Carlo Cavazzoni, Davide Ceresoli, Guido L Chiarotti, Matteo Cococcioni, Ismaila Dabo, Andrea Dal Corso, Stefano de Gironcoli, Stefano Fabris, Guido Fratesi, Ralph Gebauer, Uwe Gerstmann, Christos Gougoussis, Anton Kokalj, Michele Lazzeri, Layla Martin-Samos, Nicola Marzari, Francesco Mauri, Riccardo Mazzarello, Stefano Paolini, Alfredo Pasquarello, Lorenzo Paulatto, Carlo Sbraccia, Sandro Scandolo, Gabriele Sclauszero, Ari P Seitsonen, Alexander Smogunov, Paolo Umari, and Renata M Wentzcovitch. Quantum espresso: a modular and open-source software project for quantum simulations of materials. *Journal of Physics: Condensed Matter*, 21(39):395502, 2009.
- [2] Stefan Grimme, Jens Antony, Stephan Ehrlich, and Helge Krieg. A consistent and accurate ab initio parametrization of density functional dispersion correction (dft-d) for the 94 elements h-pu. *The Journal of Chemical Physics*, 132(15):154104, 2010.
- [3] Matteo Cococcioni and Stefano de Gironcoli. Linear response approach to the calculation of the effective interaction parameters in the LDA + U method. *Phys. Rev. B*, 71:035105, Jan 2005.

- [4] Mengchen Song, Liuting Zhang, Fuying Wu, Haoyu Zhang, Hu Zhao, Lixin Chen, and Hong Li. Recent advances of magnesium hydride as an energy storage material. *Journal of Materials Science & Technology*, 149:99–111, 2023.
- [5] Yinghui Li, Qiuyu Zhang, Li Ren, Zi Li, Xi Lin, Zhewen Ma, Haiyan Yang, Zhigang Hu, and Jianxin Zou. Core-shell nanostructured magnesium-based hydrogen storage materials: a critical review. *Ind. Chem. Mater.*, 1(3):282–298, 2023.
- [6] R. W. P. Wagemans, J. H. van Lenthe, P. E. de Jongh, A. J. van Dillen, and K. P. de Jong. Hydrogen storage in magnesium clusters: Quantum chemical study. *Journal of the American Chemical Society*, 127(47):16675–16680, 2005.